



WAVES

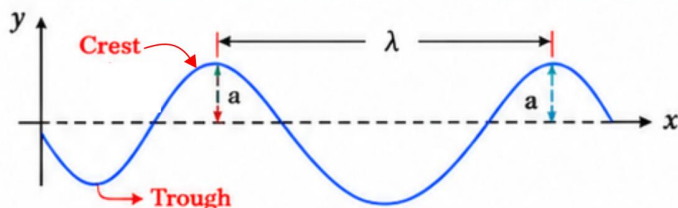


1. INTRODUCTION / WAVE IDEA

- Waves transfer energy and information from one point to another.
- In mechanical waves, particles of the medium oscillate about their mean position. The disturbance travels through the medium, not the medium as a whole.



3. DISPLACEMENT RELATION IN A PROGRESSIVE WAVE



Wave Equations

$$y(x, t) = a \sin(kx - \omega t + \phi) \rightarrow \text{Wave travelling in } +x \text{ direction}$$

$$y(x, t) = a \sin(kx + \omega t + \phi) \rightarrow \text{Wave travelling in } -x \text{ direction}$$

a = amplitude

k = angular wave number = $\frac{2\pi}{\lambda}$

ω = angular frequency = $\frac{2\pi}{T} = 2\pi\nu$

ϕ = phase constant (initial phase)

λ = wavelength

T = time period

ν = frequency = $\frac{1}{T}$

Wave Speed

$$V = \frac{\omega}{k} = \nu \lambda$$

where,

V = speed of wave

ω = angular frequency

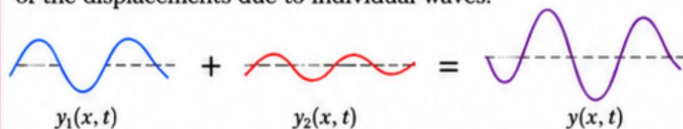
k = angular wave number

ν = frequency

λ = wavelength

5. PRINCIPLE OF SUPERPOSITION

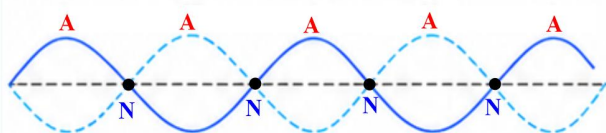
The resultant displacement at any point is the algebraic sum of the displacements due to individual waves.



7. STANDING WAVES & NORMAL MODES

Standing wave is formed by superposition of two identical waves travelling in opposite directions.

$$y = 2a \sin(kx) \cos(\omega t)$$

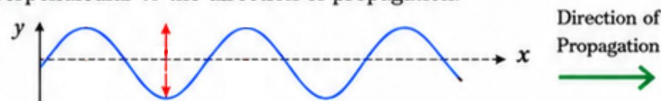


A = Antinode (maximum displacement)

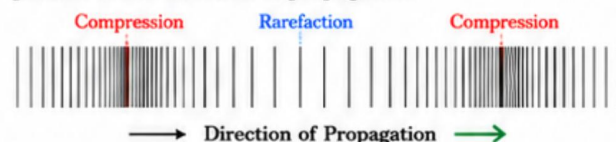
N = Node (zero displacement)

2. TYPES OF WAVES

Transverse Wave: Particle oscillation is perpendicular to the direction of propagation.



Longitudinal Wave: Particle oscillation is parallel to the direction of propagation.



4. SPEED OF A TRAVELLING WAVE

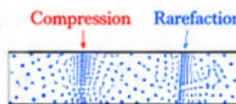
On a Stretched String



$$v = \sqrt{\frac{T}{\mu}}$$

T = tension
 μ = linear mass density

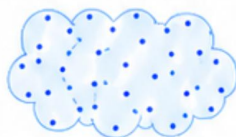
Longitudinal Wave in a Medium



$$v = \sqrt{\frac{B}{\rho}}$$

B = bulk modulus
 ρ = density

Sound in a Gas

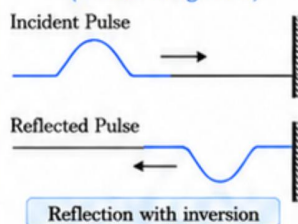


$$v = \sqrt{\frac{\gamma P}{\rho}}$$

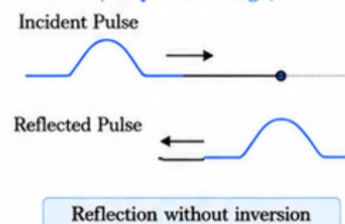
γ = ratio of specific heats
 P = pressure
 ρ = density

6. REFLECTION OF WAVES

At Rigid Boundary
(Phase change of π)



At Open / Free Boundary
(No phase change)



Normal Modes (Natural Frequencies)

String Fixed at Both Ends
or
Open Pipe (both ends open)

$$f_n = \frac{n\nu}{2L}$$

$$n = 1, 2, 3, \dots \text{ (all harmonics)}$$

$$\lambda_n = \frac{2L}{n}$$

Pipe Closed at One End
(Open at other end)

$$f_n = \frac{n\nu}{4L}$$

$$n = 1, 3, 5, \dots \text{ (odd harmonics only)}$$

$$\lambda_n = \frac{4L}{n}$$

8. BEATS

Beats arise due to interference of two sound waves of nearly equal frequencies.

Beat frequency: $f_b = |f_1 - f_2|$

Loudness varies periodically (waxing and waning).

